Magnetization and field

$\vec{M}$ = magnetic moment per unit volume

Magnetic field $\vec{B} = \mu_0 \vec{H}$

in vacuum

$\vec{B}$: magnetic induction (flux density)

$\vec{H}$: magnetic field strength (determined by free current)

$\mu_0 = 4\pi \cdot 10^{-7}\ H\,m^{-1}$ permeability of free space

in matter $\vec{B} = \mu_0(\vec{H} + \vec{M})$

$\vec{M} = \chi \vec{H}$ for linear materials

↳ SUSCEPTIBILITY

$$\vec{B} = \mu_0(1+\chi)\vec{H} = \mu_0 \mu_r \vec{H}$$

Caution: applied $\vec{B}_a, \vec{H}_a$ and internal $\vec{B}_i, \vec{H}_i$ fields are different!

$\vec{H}_i = \vec{H}_a - N\vec{M}$, N = demagnetizing factor

$\vec{B}_i = \mu_0(\vec{H}_i + \vec{M}) = \vec{B}_a + \mu_0(1-N)\vec{M}$

Ex Sphere: $N = \frac{1}{3} \Rightarrow \vec{B}_i = \vec{B}_a + \frac{2\mu_0}{3}\vec{M}$

(shows up in hyperfine coupling in atoms!)

Linear susceptibility $\chi = \frac{M}{H}$ dimensionless

intrinsic $\chi_{int} = \frac{M}{H_{int}}$

experimental $\chi_{exp} = \frac{M}{H_{applied}} = \frac{M}{H_{int} + NM} = \frac{M/H_{int}}{1 + N\, M/H_{int}} =$

$$= \frac{\chi_{int}}{1 + N\chi_{int}}$$

Note $\chi_{exp} \cong \chi_{int}$ when $\chi_{int} \ll 1$

Thermodynamics relations for magnetization $\vec{M}$

$\boxed{1}$ Maxwell's equations:

$$\begin{cases} 1)\ \vec{\nabla}\cdot\vec{D} = \rho_f & \left(\vec{D} = \varepsilon_0\vec{E} + \vec{P}\right) \\ 2)\ \vec{\nabla}\cdot\vec{B} = 0 \\ 3)\ \vec{\nabla}\times\vec{E} = -\dfrac{\partial\vec{B}}{\partial t} \\ 4)\ \vec{\nabla}\times\vec{H} = \vec{J}_f + \dfrac{\partial\vec{D}}{\partial t} & \left(\vec{H} = \dfrac{1}{\mu_0}\vec{B} - \vec{M}\right) \end{cases}$$

ρ_f = density of free charges

$\vec{J}_f$ = free current density

$$\underbrace{\vec{E}\cdot\vec{\nabla}\times\vec{H} - \vec{H}\cdot\vec{\nabla}\times\vec{E}}_{\vec{\nabla}\cdot(\vec{H}\times\vec{E})} = \vec{E}\cdot\left(\vec{J}_f + \frac{\partial\vec{D}}{\partial t}\right) + \vec{H}\cdot\frac{\partial\vec{B}}{\partial t}$$

$$= \vec{J}_f\cdot\vec{E} + \vec{E}\cdot\frac{\partial\vec{D}}{\partial t} + \vec{H}\cdot\frac{\partial\vec{B}}{\partial t}$$

Rewrite: $-\vec{J}_f\cdot\vec{E} = \vec{\nabla}\cdot(\vec{E}\times\vec{H}) + \vec{E}\cdot\frac{\partial\vec{D}}{\partial t} + \vec{H}\cdot\frac{\partial\vec{B}}{\partial t}$

Integrate both sides over the volume: $\left(\int\nabla\cdot\vec{A}\,dv = \oint\vec{A}\cdot d\vec{s}\right)$

$$-\int\vec{J}_f\cdot\vec{E}\,dV = \oint\vec{E}\times\vec{H}\cdot d\vec{s} + \int\left(\vec{E}\cdot\frac{\partial\vec{D}}{\partial t} + \vec{H}\cdot\frac{\partial\vec{B}}{\partial t}\right)dV$$

Multiply by time interval δt and define:

$$d\vec{D} = \frac{\partial\vec{D}}{\partial t}\cdot\delta t, \quad d\vec{B} = \frac{\partial\vec{B}}{\partial t}\cdot\delta t$$

$$-\int\vec{J}_f\cdot\vec{E}\,dV\,\delta t = \oint\vec{E}\times\vec{H}\cdot d\vec{S}\,\delta t + \int\left(\vec{E}\cdot d\vec{D} + \vec{H}\cdot d\vec{B}\right)dV$$

$$\vec{J}_f = nq\vec{v} \Rightarrow \vec{J}_f\cdot\vec{E} = n\vec{v}\cdot\underbrace{q\vec{E}}_{\text{force on charge } q}$$

$$\vec{J}_f\cdot\vec{E}\,\delta t = n\underbrace{q\vec{E}\cdot\vec{v}\,\delta t}_{\text{work by electric field on charge } q}$$

$$\Rightarrow -\int\vec{J}_f\cdot\vec{E}\,dV\,\delta t = \text{work done by charges} = dW$$

$$đW = \underbrace{\oint \vec{E}\times\vec{H}\cdot d\vec{S}\,\delta t}_{\text{energy emitted}} + \underbrace{\int\left(\vec{E}\cdot d\vec{D} + \vec{H}\cdot d\vec{B}\right) dV}_{\text{energy stored in the field} = đW_{em}}$$

but $\vec{E}\times\vec{H}$ = Poynting vector $\Rightarrow$ 1ST term is the energy emitted through the surface of the system during time δt. Hence the 2nd term represents energy stored in the field.

$$đW_{em} = \int\left(\vec{E}\cdot d\vec{D} + \vec{H}\cdot d\vec{B}\right) dV$$

Recall that $dF = -S\,dT - \underbrace{P\,dV}_{\text{mechanical work done on the system}}$

free energy

Thus we must extend this as (set $\vec{E} = \vec{D} = 0$ now to focus on magnetic)

$$dF = -S\,dT - p\,dV + \int \vec{H}\cdot d\vec{B}\, dV$$

But $\vec{H} = \frac{1}{\mu_0}\vec{B} - \vec{M}$

$$\Rightarrow \int dV\, \vec{H}\cdot d\vec{B} = \int dV \left\{ \underbrace{\frac{1}{2\mu_0} d(\vec{B}^2)}_{\text{energy stored in magnetic field}} - \underbrace{\vec{M}\cdot d\vec{B}}_{\text{energy due to magnetization}} \right\}$$

$$\boxed{\Rightarrow \left(\frac{\partial F}{\partial \vec{B}}\right)_{T,V} = -\int \vec{M}\cdot dV}$$

magnetization of the volume

$\boxed{2}$ Better do differently: $d\vec{B} = \mu_0\left(d\vec{H} + d\vec{M}\right)$

$$\Rightarrow \vec{H}\cdot d\vec{B} = \mu_0\, \vec{H}\cdot d\vec{H} + \mu_0\, \vec{H}\cdot d\vec{M} = \frac{\mu_0}{2} d(\vec{H}^2) + \mu_0\, \vec{H}\cdot d\vec{M}$$

Since $\vec{H}$ is determined by $\vec{J}_{free}$, the term

$\frac{\mu_0}{2}\int dV\, d(\vec{H}^2)$ represents energy stored in the field irrespective of $\vec{M}$!

Then $dF = -SdT - pdV + \int \mu_0 \vec{H} \cdot d\vec{M} \, dV$

represents free energy associated with magnetization.

Hence do legendre transformation

$$F_H = F - \int \vec{H} \cdot \vec{B} \, dV$$

$$\Rightarrow dF_H = dF - \int d\vec{H} \cdot \vec{B} \, dV - \int \vec{H} \cdot d\vec{B} \, dV$$

$$= -SdT - pdV - \int \vec{B} \cdot d\vec{H} \, dV =$$

$$= -SdT - pdV - \mu_0 \int \Big(\underbrace{\vec{H} \cdot d\vec{H} \, dV}_{\text{independent of magnetization}} + \vec{M} \cdot d\vec{H} \Big) dV$$

$$\Rightarrow \vec{M} = -\frac{1}{V\mu_0} \left(\frac{\partial F_H}{\partial H} \right)_{T,V}$$

As far as <u>linear</u> response goes, we can use F or F_H as is more convenient.

$\boxed{3}$ The same in a much simpler way (Landau, Lifshits Vol. 5, §52)

$$U = -\vec{\mu} \cdot \vec{B}$$

$$\delta U = -\vec{\mu} \cdot \delta \vec{B} \quad \Rightarrow dF = -SdT - PdV - \vec{\mu} \cdot d\vec{B}$$

$$\Rightarrow \vec{\mu} = -\left(\frac{\partial F}{\partial \vec{B}} \right)_{T,V}$$ per unit volume

Add to $\boxed{2}$

$$dF = -SdT - pdV + H\cdot dB$$

$$d\vec{B} = \mu_0\left(d\vec{H} + d\vec{M}\right)$$

$$dF = -SdT - pdV + \mu_0 H dH + \mu_0 H\cdot dM$$

let $F_H = F - \vec{H}\cdot\vec{B}$

$$dF_H = dF - dH\cdot B - H\cdot dB$$

$$= -SdT - pdV + H\cdot dB - \left(dH\cdot B + H\cdot dB\right)$$

$$= -SdT - pdV - B\cdot dH$$

$$B = \mu_0(H + M)$$

$$dF_H = -SdT - pdV - \underbrace{\mu_0\left(HdH\right.}_{\frac{\mu_0}{2}d(H^2)}\left. + MdH\right)$$

independent of magnetization

$$\Rightarrow -\left(\frac{\partial F_H}{\partial H}\right)_{T,V} = \mu_0 M$$

$\boxed{4}$

$$\hat{\mathcal{H}} = \hat{\mathcal{H}}_0 - \hat{m} \cdot \vec{B}$$

↳ magnetization operator

$$Z = \sum_{(\text{states})} e^{-E_n/k_BT} = \sum_n \langle n | e^{-\hat{\mathcal{H}}/k_BT} | n \rangle$$

Expectation value

$$\vec{M} = \langle \hat{m} \rangle = \frac{\sum_n \langle n | \hat{m} \, e^{-\hat{\mathcal{H}}/k_BT} | n \rangle}{Z} =$$

$$= \frac{\sum_n \langle n | k_BT \frac{\partial}{\partial \vec{B}} e^{-\hat{\mathcal{H}}/k_BT} | n \rangle}{Z} =$$

$$= \frac{k_BT}{Z} \frac{\partial}{\partial \vec{B}} \sum_n \langle n | e^{-\hat{\mathcal{H}}/k_BT} | n \rangle = \frac{k_BT}{Z} \frac{\partial}{\partial \vec{B}} \left(Z \right)$$

$$= k_BT \frac{\partial}{\partial \vec{B}} \left(\ln Z \right)$$

Then $TS = E - F = -\vec{M} \cdot \vec{B} + k_BT \ln Z$

At fixed T : $T\,dS = -\vec{M} \cdot d\vec{B} - d\vec{M} \cdot \vec{B} + \underbrace{k_BT \frac{\partial \ln Z}{\partial \vec{B}}}_{\vec{M}} \cdot d\vec{B} =$

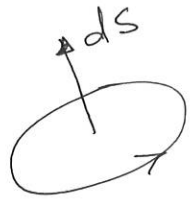
$$= -\vec{B} \cdot d\vec{M}$$

$$\Rightarrow dF = dE - T\,dS = d(-\vec{M} \cdot \vec{B}) + \vec{B} \cdot d\vec{M} =$$

$$= -\vec{M} \cdot d\vec{B} \quad (\text{at } T = \text{const})$$

$\Rightarrow$ again $\vec{M} = -\left(\frac{\partial F}{\partial \vec{B}} \right)_T$

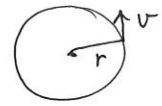
Magnetic moment



$$d\vec{\mu} = I\, d\vec{S}$$

$$\vec{\mu} = \int d\vec{\mu} = I \int d\vec{S}$$

Orbital angular momentum $|\vec{L}| = m v r$



$$I = \frac{-e}{\tau}, \quad \tau = \frac{2\pi r}{v}$$

$$\mu = I \pi r^2 = \frac{-ev}{2\pi r} \cdot \pi r^2 = \frac{-e}{2} v r = -\frac{e}{2m} L$$

Generally, $\vec{\mu} = \gamma \vec{L}$; γ = gyromagnetic ratio

$$\gamma_{orbital} = \frac{q}{2m}$$

Quantum Mechanics: $L_z |\psi\rangle = \hbar (\text{integer}) |\psi\rangle \Rightarrow \mu_z = \frac{-e\hbar}{2m} (\text{integer})$

$\Rightarrow$ magnetic moment is quantized in units of

$$\mu_B = -\frac{e\hbar}{2m} \quad \text{(Bohr magneton)}$$

Precession

energy $\quad E = -\vec{\mu} \cdot \vec{B}$

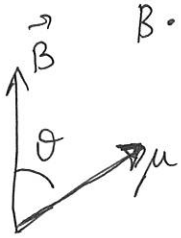
torque $\quad \vec{G} = \vec{\mu} \times \vec{B}$

but $\frac{d}{dt}\vec{L} = \vec{G} \Rightarrow \frac{d}{dt}\frac{\vec{\mu}}{\gamma} = \vec{\mu} \times \vec{B}$

$$\boxed{\frac{d\vec{\mu}}{dt} = \gamma\, \vec{\mu} \times \vec{B}}$$

Observe that $\frac{d}{dt}(\vec{\mu}^2) = 2\vec{\mu} \cdot \frac{d\vec{\mu}}{dt} = 2\gamma\, \vec{\mu} \cdot \vec{\mu} \times \vec{B} = 0$

$$\vec{B} \cdot \frac{d}{dt}\vec{\mu} = \frac{d}{dt}(\vec{\mu} \cdot \vec{B}) = \gamma\, \vec{B} \cdot \vec{\mu} \times \vec{B} = 0$$

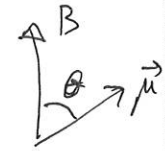


$\Rightarrow$ angle θ is conserved.

$|\vec{\mu}|$ is conserved.

Example let $\vec{B} = (0,0,B)$ $\quad \vec{\mu} = (\mu_x, \mu_y, \mu_z)$

$$\vec{G} = \vec{\mu} \times \vec{B} = \begin{vmatrix} \hat{x} & \hat{y} & \hat{z} \\ \mu_x & \mu_y & \mu_z \\ 0 & 0 & B \end{vmatrix} = \hat{x}\,\mu_y B - \hat{y}\,\mu_x B$$



$$\Rightarrow \left.\begin{aligned} \frac{d}{dt}\mu_x &= \gamma B \mu_y \\ \frac{d}{dt}\mu_y &= -\gamma B \mu_x \end{aligned}\right\} \quad \begin{aligned} &\frac{d^2\mu_x}{dt^2} = -\gamma^2 B^2 \mu_x \\ &\mu_x(t) = \underbrace{\mu_x(0)}_{|\vec{\mu}|\sin\theta} \cos(\gamma B t) \end{aligned}$$

$$\begin{cases} \mu_x(t) = |\vec{\mu}| \sin\theta \cos(\omega_L t) \\ \mu_y(t) = |\vec{\mu}| \sin\theta \sin(\omega_L t) \\ \mu_z(t) = |\vec{\mu}| \cos\theta \end{cases} \qquad \omega_L = \gamma B \quad \text{Larmor frequency}$$

[Bohr magneton $\mu_B = \dfrac{-e\hbar}{2m}$
$L = n\hbar$]

Bohr-van Leeuwen Theorem

$$\vec{p} = m\vec{v} + q\vec{A} \qquad \vec{B} = \vec{\nabla} \times \vec{A}, \quad \vec{E} = -\vec{\nabla}\varphi - \frac{\partial \vec{A}}{\partial t}$$

$$\cancel{H_0 = \frac{m\vec{v}^2}{2m}} \quad K.E. = \frac{1}{2} m \vec{v}^2 \Rightarrow \frac{(\vec{p} - q\vec{A})^2}{2m} = \frac{(-i\hbar\vec{\nabla} - q\vec{A})^2}{2m}$$

System of N-particles

$$E = \sum_i \frac{1}{2m}(\vec{p}_i - q\vec{A}_i)^2 + V(\{\vec{r}\})$$

Magnetization $\vec{\mu} = -\left(\dfrac{\partial E}{\partial B}\right)_{T,V}$, $\quad F = E - TS = -k_B T \ln Z$

$$\Rightarrow \vec{\mu} = k_B T \frac{\partial}{\partial B} \ln Z$$

$$\beta = 1/k_B T$$

$$Z(A) = \int \prod d\vec{p}_i \, d\vec{r}_i \; e^{-\beta E} = \int \prod_i d\vec{p}_i \, d\vec{r}_i \; e^{-\beta \left[\sum \frac{(p_i - q A_i)^2}{2m} + V \right]}$$

$$= \left(\text{shift } \vec{p}_i - q\vec{A}_i = \vec{q}_i \right) = \int \prod d\vec{q}_i \, d\vec{r}_i \; e^{-\beta \left[\sum \frac{\vec{q}_i^{\,2}}{2m} + V \right]} = Z(0)$$

$\Rightarrow$ Z does not depend on $\vec{B}$

$$\Rightarrow \vec{\mu} = (k_B T) \frac{\partial}{\partial B} \ln Z(B) = 0$$

$$\boxed{\text{Purely classical system does not have magnetization even in the presence of magnetic field}}$$

$\Rightarrow$ any magnetism is <u>QUANTUM</u>